
Machine Learning Outlier Detection for Robust Causal Inference

How outliers bias treatment effects, and a practical fix

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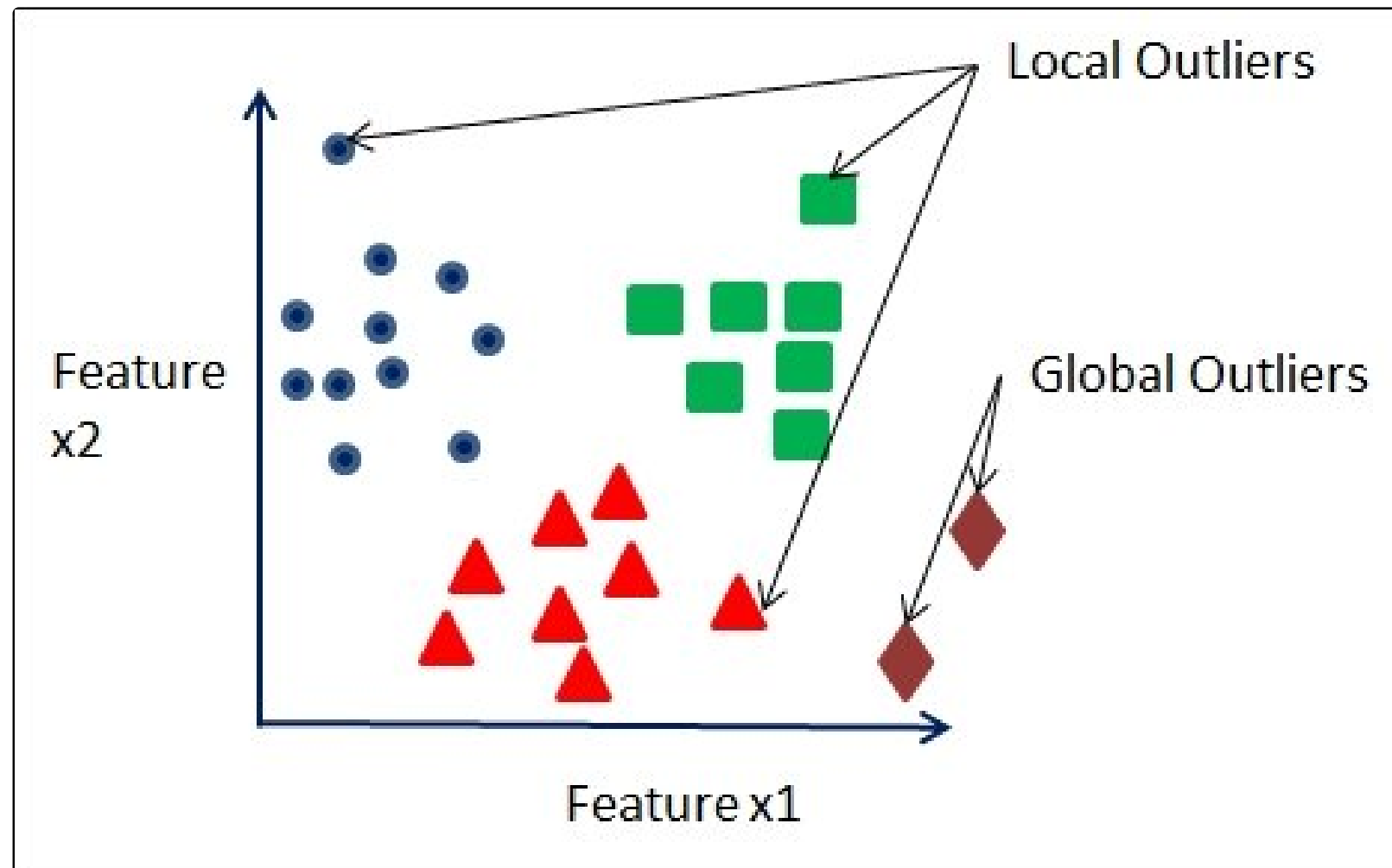
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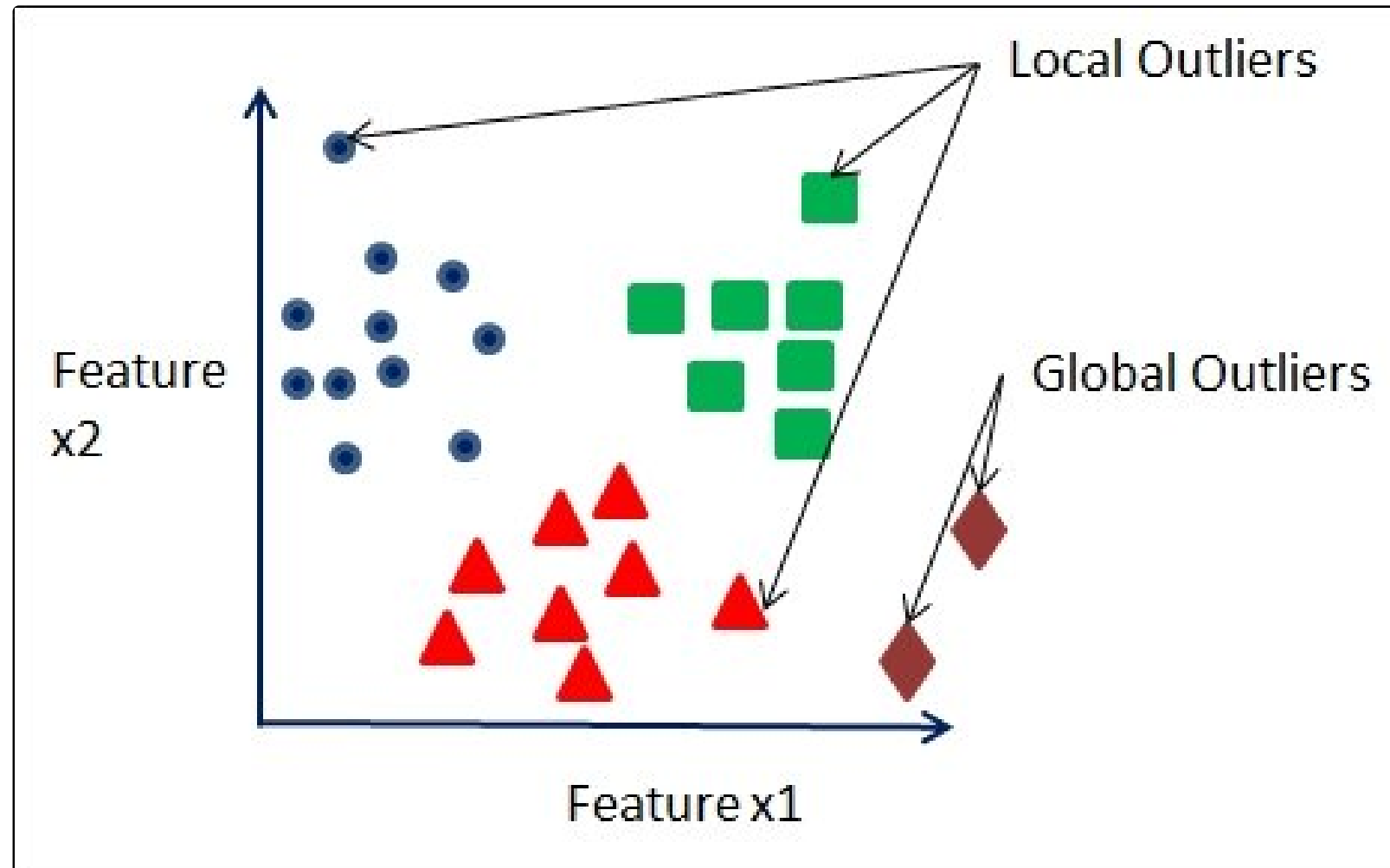
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$$Y = D\beta + g(X) + \varepsilon \implies \tilde{Y} = \tilde{D}\beta + \tilde{\varepsilon}$$

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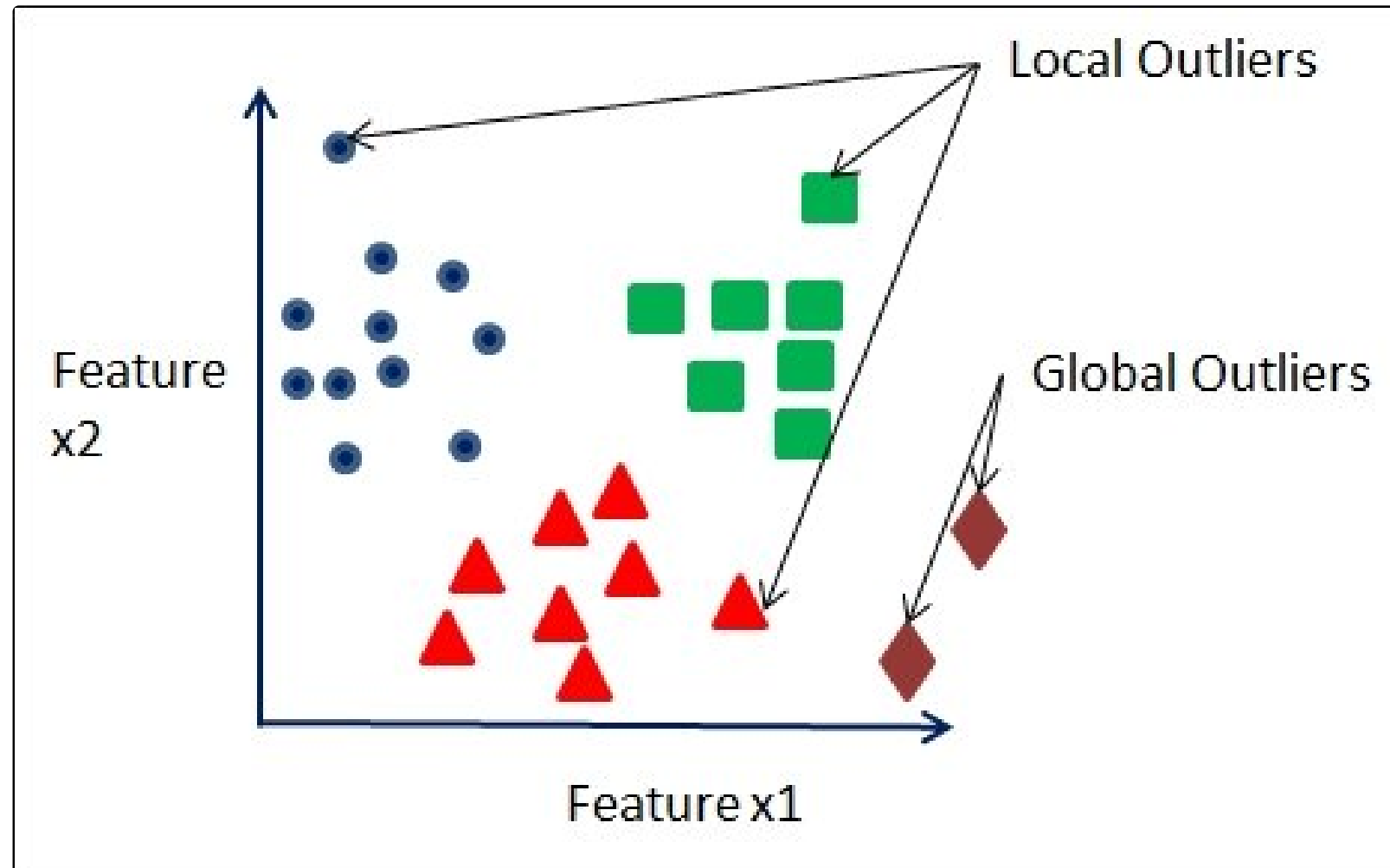


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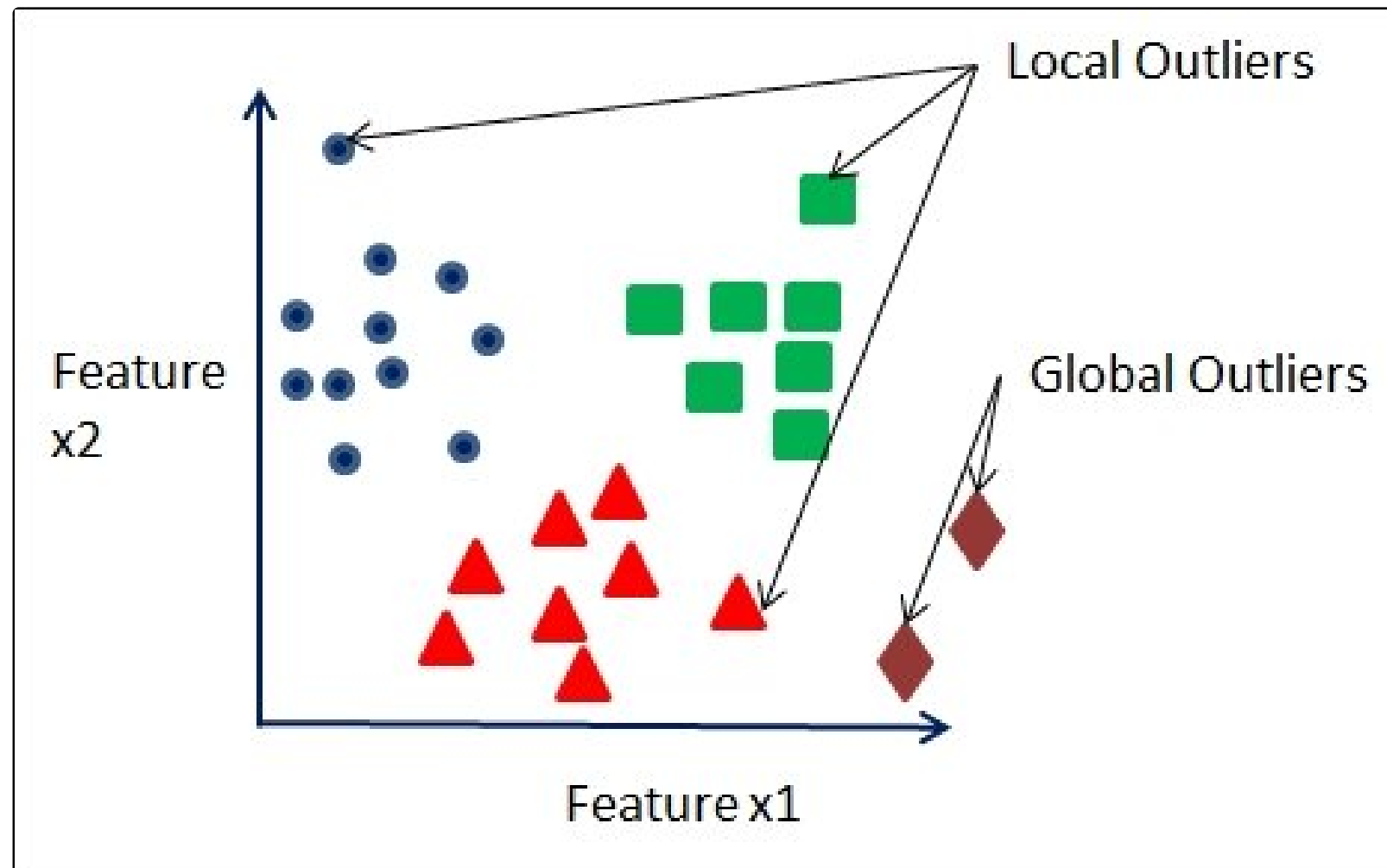
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- Evaluate 6 **multivariate, unsupervised** ML outlier methods from PyOD:
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- Detect both **local** (between clusters) and **global** (extreme) outliers
- Propose an **ensemble**: only remove observations *all models agree* are outliers

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No removal	0%	-0.01	[-0.02, 0.00]
Ensemble	1%	17.55	[16.68, 18.42]
Percentile 95th	98%	28.47	[18.93, 38.01]

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1% outliers drove the ATE from 17.43 → 0. The ensemble recovers the truth.

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Percentile 95th	35.8%	0.26	[−1.70, 2.22]	No

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A "significant" effect of 4.44 disappears, removing just 0.8% of data.

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Thank you! Questions?